



CSA0501-DATABASE MANAGEMENT SYSTEMS

LAB EXPERIMENTS

DATE: 6/08/2026

EXPERIMENT 16 -- HIGH LEVEL PROGRAMMING EXTENSIONS - PROCEDURES

AIM:

To implement **stored procedures** using MySQL for retrieving and processing records from the student_cgpa table.

PROCEDURE:

1. Select the database containing the student_cgpa table.
2. Create a stored procedure using CREATE PROCEDURE.
3. Use the procedure to retrieve records from the table.
4. Create another procedure that accepts a student's register number as input.
5. Retrieve the student's CGPA.
6. Determine the student's performance level based on CGPA.
7. Execute the procedures using the CALL statement.
8. Display the results.

PROGRAM 1:

Create a simple procedure to get all the records from the student_cgpa table.

1. CREATE TABLE

```
CREATE TABLE student_cgpa (  
  reg_no VARCHAR(15) PRIMARY KEY,  
  student_name VARCHAR(50) NOT NULL,  
  department VARCHAR(20),  
  no_of_subs_pass INT,  
  no_of_subs_fail INT,  
  cgpa DECIMAL(3,2)  
);
```

Then Insert Values:

```
INSERT INTO student_cgpa  
(reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa)  
VALUES  
(192411123,'Deepthi','CSE',22,2,8.71),  
(192512121,'Varun','IT',27,1,9.23),  
(192312117,'Aadhya','ECE',25,3,8.97),
```

```
( '192411227','Dimple','CSE',31,0,9.37),
('192820896','Mohan','EEE',18,9,6.48),
('192411122','Yogitha','CSE',31,1,9.01),
( '192516871','Sadhya','IT',25,5,7.22),
('192111101','Rahul','MECH',30,12,5.14),
('192411010','Alekhyia','CSE',35,2,9.01),
('192511135','Rahul','CIVIL',37,0,9.72);
```

```
SELECT * FROM student_cgpa;
```

OUTPUT:

reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhyia	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192820896	Mohan	EEE	18	9	6.48

#	Time	Action	Message	Duration / Fetch
1	22:41:47	CREATE TABLE student_cgpa (reg_no VARCHAR(15) PRIMARY KEY, student_name VARCHAR(50) NO...	0 row(s) affected	0.031 sec
2	22:44:31	INSERT INTO student_cgpa (reg_no, student_name, department, no_of_subs_pass, no_of_subs_fail, cgpa) VA...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
3	22:46:18	SELECT * FROM student_cgpa LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

2. PROGRAM

```
DELIMITER //
```

```
CREATE PROCEDURE GetStudentDetails()
BEGIN
    SELECT * FROM student_cgpa;
END //
```

```
DELIMITER ;
CALL GetStudentDetails();
```

OUTPUT:

Result Grid		Filter Rows:		Export: 	Wrap Cell Content: 
reg_no	student_name	department	no_of_subs_pass	no_of_subs_fail	cgpa
192111101	Rahul	MECH	30	12	5.14
192312117	Aadhya	ECE	25	3	8.97
192411010	Alekhya	CSE	35	2	9.01
192411122	Yogitha	CSE	31	1	9.01
192411123	Deepthi	CSE	22	2	8.71
192411227	Dimple	CSE	31	0	9.37
192511135	Rahul	CIVIL	37	0	9.72
192512121	Varun	IT	27	1	9.23
192516871	Sadhya	IT	25	5	7.22
192611200	Kavya	CSE	28	1	8.65
192820896	Mohan	EEE	18	9	6.48

result 17 ×

PROGRAM 2.

Give the user permission to SELECT data from your student_cgpa table.

Create a stored procedure that accepts a **register number** and determines the student's performance level based on CGPA.

- **CGPA** $\geq$ **9.00** → EXCELLENT
- **CGPA** $\geq$ **8.00** → GOOD
- **CGPA** $<$ **8.00** → AVERAGE

PROGRAM:

```

DELIMITER //

CREATE PROCEDURE GetStudentLevel(
    IN p_reg_no INT,
    OUT p_level VARCHAR(20)
)
BEGIN
    DECLARE student_cgpa DECIMAL(3,2);

    SELECT cgpa
    INTO student_cgpa
    FROM student_cgpa
    WHERE reg_no = p_reg_no;

    IF student_cgpa >= 9.00 THEN
        SET p_level = 'EXCELLENT';

    ELSEIF student_cgpa >= 8.00 THEN
        SET p_level = 'GOOD';

    ELSE
        SET p_level = 'AVERAGE';
    END IF;
END IF;

```

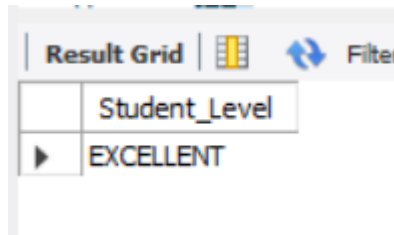
```
END //

DELIMITER ;

CALL GetStudentLevel(192512121, @level);

SELECT @level AS Student_Level;
```

OUTPUT:



The screenshot shows a 'Result Grid' window with a single column header 'Student_Level' and one data row containing the value 'EXCELLENT'. The window has a 'Filter' button and a grid icon in the top right corner.

Student_Level
EXCELLENT

RESULT: Thus the high level procedures programs are executed successfully.